

Transforming the chemotherapy of human African trypanosomiasis

Michael P. Barrett¹

AUTHOR AFFILIATION See affiliation list on p. 18.

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SUMMARY Prior to 2019, when the orally available drug fexinidazole began its clinical use, the treatment of human African trypanosomiasis (HAT) was complex and unsatisfactory for many reasons. Two sub-species of the *Trypanosoma brucei* parasite are responsible for HAT, namely the rhodesiense form found in East and Southern Africa and the gambiense form found in Central and West Africa. Diseases caused by both forms manifest in two stages: stage 1 before and stage 2 after central nervous system involvement. Prior to 2019, different drugs were required for each of the two parasite sub-species at each stage. Gambiense disease required pentamidine or nifurtimox-eflornithine combination therapy, while for rhodesiense disease, suramin or melarsoprol was given for stages 1 and 2, respectively. These drugs all suffered complications including protracted administration regimens involving multiple injections with drug-induced

Editor Louisa A. Messenger, University of Nevada Las Vegas, Las Vegas, Nevada, USA

Address correspondence to Michael P. Barrett, michael.barrett@glasgow.ac.uk.

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adverse effects common. Today, a single drug, fexinidazole, can be given orally in most cases for both diseases at either stage. Another compound, acoziborole, effective in both stages 1 and 2 gambiense disease with a single dosing is anticipated to become available within a few years. Moreover, the recent engagement of multilateral organizations in seeking other compounds that could be used in HAT therapy has also been successful, and a rich vein of new trypanocides has been discovered. Here, the clinical use, modes of action, and resistance risks for drugs used against HAT are discussed.

KEYWORDS African trypanosomiasis, chemotherapy, mode of action, resistance, fexinidazole, suramin, pentamidine, acoziborole, nifurtimoc-eflornithine combination therapy, melarsoprol

INTRODUCTION

Background

Human African trypanosomiasis (HAT) is commonly referred to as sleeping sickness given that among neurological symptoms associated with the second stage of the disease, when parasites have entered and provoked changes to the central nervous system, altered sleep–wake patterns are common (1–4). HAT is endemic to sub-Saharan Africa. Two sub-species of *Trypanosoma brucei* are responsible for HAT. *T. b. gambiense* causes a chronic disease in West and Central Africa, while *T. b. rhodesiense* causes a relatively acute disease in Eastern and Southern Africa (1–4). The biological differences between the parasites which engender the difference in disease progression are unknown. However, both avoid destruction by an innate immune mechanism, mediated by trypanosome lytic factors (TLFs) that include the lipoprotein ApoL1 (4, 5), which enters in complexes with other lipoproteins and haptoglobin via a parasite haptoglobin–hemoglobin receptor (TbHpHbR) (6). *T. b. rhodesiense* has a serum resistance-associated (*sra*) gene that is a relic variant surface glycoprotein VSG gene (7), the product of which binds ApoL1 when it is expressed (VSGs are described in more detail below). *T. b. gambiense* has a different gene, *Tgsgp* (8), also a relic mutated *vsg*, which encodes a protein TgsGP that, along with other mechanisms, also enables these parasites to avoid the trypanosome lytic factor effects (9, 10). In addition to their ability to withstand anti-trypanosomal innate immunity, the parasites avoid destruction by antibody or complement-mediated lysis by virtue of a variant surface glycoprotein coat that shields the surface from these immunological effectors (11–13). When antibodies specific to one coat are produced, they do lead to the death of those trypanosomes expressing that particular coat. However, *T. brucei* has in the order of 2,000 different coat protein-encoding *vsg* genes available (14), and they can switch, more or less *ad infinitum*, to yield new variants not recognized by antibodies produced by previous variants. Other trypanosome species cause African animal trypanosomosis (15) but are not infectious to humans due to the trypanosome lytic factors. In very rare instances, for example, in humans defective in trypanosome lytic factors, infection with species that are not normally human infectious, particularly *Trypanosoma evansi*, can ensue (16). In Latin America, another, more distantly related trypanosome, *Trypanosoma cruzi*, is responsible for Chagas disease (17). This review will focus on human African trypanosomiasis caused by *T. b. gambiense* and *T. b. rhodesiense*.

The parasites are transmitted by tsetse flies of the genus *Glossina*, of which both males and females feed on blood and can transmit parasites that take up residence in the tsetse midgut following feeding upon a mammalian host (18). The focal distribution of HAT is due to complex interactions among the parasite, tsetse vector, host, and environment (1–4). The disease typically occurs in rural areas, where human contact with tsetse is common but can also affect peri-urban regions. Activities such as farming, hunting, fishing, and collecting water and wood expose people to tsetse, putting people of either sex and at all age groups at risk, though adults are more commonly affected than children.

The gambiense parasite is primarily associated with man, and its epidemiology is essentially anthroponotic. *T. b. rhodesiense*, on the other hand, can also infect a range of wild and domestic animals and appears very closely related to *T. b. brucei* which is not infectious to humans due to TLFs (noted above).

Following inoculation from a tsetse fly, an incubation period of variable time ensues followed by stage 1 (hemolymphatic) disease where parasites multiply in the blood and lymphatic system (1–4). Symptoms in stage 1 are non-specific, e.g., swollen lymph nodes, headache, fever, weakness, joint, and muscle pain. Over time, some parasites can cross the blood–brain barrier and induce changes to the central nervous system initiating stage 2 (meningoencephalitic) disease. Here, the disease is characterized by neurological issues such as sleep disturbances (hence “sleeping sickness”), sensory changes, abnormal muscle tone, ataxia, psychiatric symptoms, seizures, coma, and eventually death.

Drugs exist to treat HAT, and the history of chemotherapy itself is entwined with trypanosomiasis chemotherapy, particularly through the early work of Paul Ehrlich (19). His work on arsenicals and antimicrobial dyes led directly to the arsenical drug salvarsen and arsenic continued to be used in later drugs like tryparsamide and then melarsoprol. Trypanocidal dyes, including trypan blue, underpinned developments leading to the naphthalene derivative suramin (20).

The initial recorded epidemics of HAT coincided with the period during which European nations sought to colonize large swathes of Africa, and the disease has been referred to as “the colonial disease” (21). By the time many African nations were gaining independence, the disease was in decline. However, as surveillance waned, a major resurgence occurred late in the 20th century (22). This precipitated enhanced activity to combat the disease (23). The World Health Organization (WHO) has coordinated efforts to bring HAT under control and the Drugs for Neglected Diseases initiative (DNDi), a Geneva based not-for-profit Product Development Partnership, that spun out from Médecins Sans Frontières took on HAT as a top priority (24). The sustained efforts of National Sleeping Sickness Control Programmes (NSSCPs) over the past 25 years have drastically reduced the incidence of HAT, with fewer than 1,000 cases reported annually since 2018 (25). WHO aims to eliminate HAT, targeting the elimination of *T. b. gambiense* transmission by 2030 and keeping *T. b. rhodesiense* HAT below the threshold for consideration as a public health problem by the same date.

Because of the complications and risks associated with stage 2 drugs over stage 1 drugs, and some differences in biochemistry between the two parasites, the different types of HAT and different stages traditionally required separate therapies. Prior to the WHO interim guidelines issued in 2019 (26), the treatment for gambiense HAT included eflornithine, melarsoprol, nifurtimox (with eflornithine), and pentamidine. For rhodesiense HAT, melarsoprol was recommended for stage 2 and suramin for stage 1. These medications are all donated by manufacturers to WHO. Sanofi currently provides fexinidazole, eflornithine, pentamidine, and melarsoprol, while Bayer provides suramin and nifurtimox. WHO then controls the distribution of drugs to affected areas to ensure access to treatment for all HAT patients. New guidelines for treatment were issued in 2024 ([27]; <https://www.who.int/publications/i/item/9789240096035>; Table 1).

Diagnosis and staging of HAT

Clinical signs and symptoms, as outlined above, in patients in HAT endemic localities, can give suspicion of HAT but are inadequate for a confirmed diagnosis (1–3). Antibody tests exist for screening populations at risk of *T. b. gambiense* infection, but parasitological confirmation by observing trypanosomes in body fluids by microscopy has classically been necessary because of the complexity and potential risks of chemotherapy (28). Serological tests have been very useful for mass screening for gambiense HAT, the card agglutination test for trypanosomiasis remaining the most sensitive (29), although simplified lateral flow tests have been developed in more recent years (30, 31). Serological tests have not yet been developed for rhodesiense HAT, as the relative ease of finding parasites in blood samples in patients renders them less necessary, although

TABLE 1 Overview of WHO treatment recommendations for gambiense HAT and rhodesiense HAT prior to 2019 and today^a

Disease and stage	Treatment	
	Pre-2019	Current recommendations (2024)
Gambiense HAT		
Stage 1	Pentamidine (i.m.)	Fexinidazole (oral)
Stage 2 (early)	Nifurtimox–eflornithine combination (oral and i.v.)	Fexinidazole (oral)
Stage 2 (late)	Nifurtimox–eflornithine combination (oral and i.v.)	Nifurtimox–eflornithine combination (oral and i.v.)
Rhodesiense HAT		
Stage 1	Suramin (i.v.)	Fexinidazole (oral)
Stage 2	Melarsoprol (i.v.)	Fexinidazole (oral)

^aThe drugs used for HAT prior to WHO interim guidelines of 2019, and those following a guideline review in 2024 are listed for both *T. b. gambiense* and *T. b. rhodesiense* disease at stages 1 and 2. Routes of administration are noted as i.m. (intramuscular), i.v. (intravenous), and oral.

advances in rhodesiense-HAT diagnosis are being made (32). In the case of stage 1 gambiense disease, parasite numbers in blood are frequently very low, and it is necessary to first enrich for trypanosomes, for example, using mini-anion exchange chromatography (33). For stage 2 disease, parasites may also be below microscopy detection limits in cerebrospinal fluid (CSF), in which case finding white cells in CSF (>5/μL) in confirmed cases is also considered adequate to conclude that the disease has reached stage 2. As gambiense disease progresses, the white cell count also increases, and in recent years, a further distinction between an early stage 2 infection where white blood cells (WBC) is below 100/μL and late stage 2 where more than 100 white cells/μL are found determines whether therapy should be with fexinidazole or nifurtimox–eflornithine combination therapy (NECT) (27) which is outlined later.

Molecular tests seeking parasite nucleic acids, e.g., through PCR, LAMP (34), or the CRISPR-based SHERLOCK approach (35), have offered the possibility of improved sensitivity, although standardized approaches for routine field use have yet to emerge.

Advances against HAT in the 21st century have been remarkable. New WHO guidelines, released in 2024 (27), which now recommend fexinidazole as the first-line treatment for both gambiense and rhodesiense HAT, mark this success, and the future availability of acoziborole (36) as a single-dose oral medicine points to hope that HAT elimination is a genuine possibility.

FEXINIDAZOLE

Background

Fexinidazole (Fig. 1) is a 2-substituted 5-nitroimidazole developed by Hoescht as Hoe 239 in the 1970s (37). Its trypanocidal activity was shown by Jennings et al. in the 1980s (38), but its development was arrested over general concerns relating to genotoxicity issues associated with other nitroheterocyclic compounds. Jennings and coworkers showed that 250 mg/kg/day over 4 consecutive days cured 80% of mice infected with the GVR35 *T. brucei* model of stage 2 disease (38), but it was necessary to include suramin to reach a cure of 100% of mice. In the late 1990s, interest in nitroimidazoles was rekindled when a compound of this class, megazol, showed high trypanocidal potency (39). Megazol, however, was highly genotoxic and thus discontinued. However, when DNDi began a quest for novel trypanocides, nitroimidazoles were on the agenda. DNDi collected over 800 nitroheterocyclic compounds from various sources and screened them through the Swiss Tropical and Public Health Institute against *T. b. rhodesiense* (40). Eighty molecules were trypanocidal, of which 35 showed a high selectivity index (SI >100) when compared to a counter screen against L6 rat fibroblasts. Eight compounds showed effectiveness in curing a mouse model of stage 1 disease with oral dosing, while four compounds,

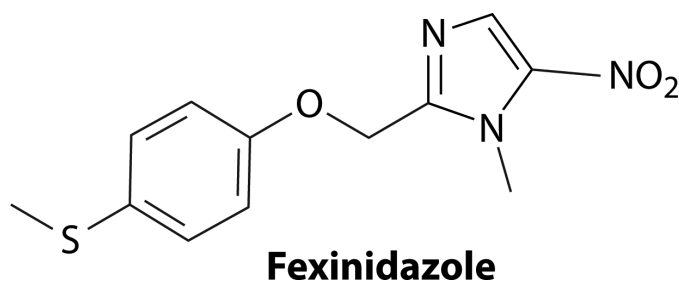


FIG 1 Structure of fexinidazole.

including fexinidazole, were also effective in curing the mouse model of stage 2 central nervous system (CNS)-involved disease (41).

Fexinidazole is metabolized to two primary metabolites: fexinidazole-sulfoxide (M1) and fexinidazole-sulfone (M2) (42), both of which retain activity against trypanosomes with *in vitro* measured IC_{50} values ranging from 0.7 to 3.3 μ M depending on the trypanosome strain tested (43). Oral administration of fexinidazole at 100 mg/kg/day for 4 days successfully cured acute mouse models infected with *T. b. rhodesiense* (STIB900) or *T. b. gambiense* (130R), while the *T. b. brucei* GVR35 stage 2 model required 100 mg/kg twice daily for 5 days to cure (43).

Fexinidazole was shown to have oral bioavailability and to distribute well throughout the body, including entry to the brain (41). Pharmacological and toxicological studies confirmed that fexinidazole is safe and well tolerated, opening the way to clinical development (44, 45).

Clinical development and use

Following phase 1 trials, fexinidazole's safety and efficacy against gambiense HAT were evaluated in a pivotal phase II/III trial involving stage 2 patients in the Democratic Republic of Congo (DRC) and the Central African Republic (45), along with associated studies covering stage 1 and early stage 2 patients and children aged 6–15 years across all disease stages (46).

Fexinidazole proved effective in these trials, leading to a positive scientific opinion by the European Medicines Agency in 2018 (47) and the inclusion of fexinidazole against gambiense HAT in new WHO interim guidelines in 2019 (26). Countries affected by HAT, such as DRC and Uganda, approved and implemented the new treatment. The United States Food and Drug Administration (FDA) granted its approval in 2021 (48).

Sanofi has committed to donating fexinidazole to the WHO, which then distributes it to the NSSCPs of HAT-endemic countries, making the treatment accessible and free of charge.

Marketed as Fexinidazole Winthrop, the drug is available as pale-yellow tablets, each containing 600 mg of active ingredient. The tablets are packaged in wallets: one type with 14 tablets for children aged 6 years and older and people weighing 20–34 kg, the other type with 24 tablets for adults weighing 35 kg or more.

As an oral therapy, fexinidazole has advantages over medicines requiring injection, which carry risks (e.g., catheter or needle-related infections) and require hospitalization. Oral administration also reduces health-system-related costs by eliminating the need for additional materials and logistics, and some patients can receive treatment closer to home.

In patients weighing ≥ 35 kg fexinidazole is taken once daily for 10 days, with a loading dose (1,800 mg; three tablets) for the first 4 days and a maintenance dose (1,200 mg; two tablets) for the remaining 6 days. Tablets should be taken whole (although investigations into using them crushed are in progress) and with food to assure proper absorption. Medical supervision is required for the dosing to assure the correct administration. In patients ≥ 6 years old and a weight of 20–34 kg, a loading dose

of 1,200 mg (two tablets) for 4 days then a single 600 mg tablet for the remaining 6 days is given.

The drug can be given to elderly patients with the same regimen, but consideration needs to be given in certain circumstances, e.g., in patients with gastrointestinal (GI) issues or hepatic impairment.

Fexinidazole should not be used in children below 6 years old or anyone weighing below 20 kg. It is also contraindicated in people with hypersensitivity to the drug itself or other nitroimidazoles. Individuals with hepatic impairment, a risk of QT interval prolongation due to congenital conditions, uncorrected electrolyte imbalances, history of cardiac arrhythmia, severe congestive heart failure, or a family history of sudden death should also not receive the drug.

Common adverse reactions include nausea, vomiting, decreased appetite, headache, insomnia, tremor, and dizziness. Neuropsychiatric reactions, including suicidal ideation, have been reported; hence, caution is advised for patients with psychiatric disorders. QT interval prolongation and neutropenia may also occur, requiring careful monitoring. Alcohol should be avoided as the drug is a nitroimidazole, and there is a concern that similar reactions to those observed with metronidazole could occur (27).

Mode of action and resistance mechanisms

Fexinidazole is a prodrug. Within the mammalian host, it is rapidly metabolized to sulfoxide and sulfone derivatives, which share similar trypanocidal activity to the parent (42). Upon entry into trypanosomes, the drug is further metabolized. For example, a mitochondrial nitroreductase enzyme (TbNTR1) (49) transfers electrons to the compound's nitrogroup, creating highly reactive molecules believed to bind to a plethora of host cellular molecules (proteins, metabolites, nucleic acids, lipids, etc.), leading to multi-system failure and death, with inhibition of DNA synthesis recently shown to be an early consequence of fexinidazole in trypanosomes (50). The same nitroreductase enzyme, which is found in trypanosomatids (and orthologous enzymes are found in many bacteria), has no analog in mammalian cells and is also responsible for activating other trypanocidal nitroheterocycles including nifurtimox (51) and benznidazole (52) (two compounds used in Chagas disease therapy caused by *T. cruzi* as well as HAT in the case of nifurtimox). Of potential importance, NTR1 is distinct from the pyruvate-ferredoxin oxidoreductase system, which is operative in anaerobic protists like *Entamoeba*, *Giardia*, and *Trichomonas* (53), and plays a role in the reduction of other nitroheterocyclic compounds (e.g., metronidazole) in these organisms. Notably, metronidazole has only weak activity against trypanosomes, as it is a poor substrate for TbNTR1 (52). This makes it unlikely that undiagnosed trypanosomes in a patient treated for a separate anaerobe infection with metronidazole would develop resistance to fexinidazole, although the possibility exists, theoretically, that selection of resistance to nifurtimox (used in NECT) could lead to mutations in TbNTR1 and consequent cross-resistance to fexinidazole (54). One copy of the *tbnt1* gene (that encodes NTR1), along with a large stretch of adjacent genome sequence, was lost in trypanosomes selected for resistance to nifurtimox (54). In the same study, the selection of resistance to fexinidazole was associated with a reduction in copy number of DNA flanking *tbnt1* in one allele, without diminishing *tbnt1* gene copy number itself. The level of mRNA transcript of the gene was, however, halved. Knocking out of a single copy of the two genes (trypanosomes are diploid and therefore have duplicates of most genes across their paired chromosomes) gives only low-level resistance to fexinidazole; hence, other genes must play a role in activity and resistance to the drug (54).

Fexinidazole has also been shown to have activity against *T. cruzi* (55) and *Leishmania* parasites (56), but potency has been inadequate to proceed to late-phase clinical trials as a monotherapy, although use in combinations remains a possibility for the diseases caused by these parasites.

PENTAMIDINE

Background

Pentamidine (Fig. 2) was, for many years, the drug of choice against stage 1 gambiense HAT. It is now the preferred treatment only in children under 6 years old or in patients weighing less than 20 kg, as well as other stage 1 gambiense HAT patients for whom fexinidazole is not appropriate (27). It may also be used in stage 1 rhodesiense disease if other drugs (e.g., fexinidazole or suramin) are not available. The drug is usually administered via deep intramuscular injection due to the risk of severe hypotension with intravenous delivery (57). It is also a second-line treatment for *Pneumocystis jirovecii* pneumonia and occasionally used in leishmaniasis therapy (58). The development of pentamidine was an early example where rational approaches to drug design yielded successful compounds, albeit for reasons other than the initial rationale. Bloodstream-form trypanosomes are prodigious users of glucose and entirely dependent upon it as a carbon and energy source (59). It was known that trypanosome infections of rodents rapidly led to hypoglycemia, and thus, the ability of glucose-lowering agents, including insulin (which had little impact) and synthalin, a synthetic analog of insulin, which had a significant impact was tested (60, 61). Synthalin is a bis-guanidine dication, and this prompted the synthesis of a series of other dications, including diamidines such as stilbamidine and pentamidine, which displayed exceptional potency against the parasites (62), independent of any host effect, ruling out the hypothesized role for diminished host glucose. Pentamidine had a better safety profile and hence was developed further. Development of diamidines continued into the 21st century, and the synthesis of an orally available prodrug version of furamidine, known as pafuramidine, entered clinical trials against stage 1 gambiense HAT (62, 63). Unfortunately, renal toxicity appeared in several subjects in a retrospective phase 1 safety trial which led to the compound being abandoned (64), in spite of its having shown good efficacy in a phase II trial of 273 stage 1 patients across six sites in DRC, Angola, and South Sudan (63). The same program also identified other, closely related structures that could enter the CSF and cure mouse models of stage II disease (65, 66). However, the diamidine program closed following the abandonment of pafuramidine, and the emergence of fexinidazole and later acoziborole meant the series was put on indefinite hold.

Clinical use

Pentamidine is manufactured by Sanofi and is available in vials containing 300 mg of pentamidine isethionate powder, which must be reconstituted with 10 mL of sterile water for injection (27). The dosage is 4 mg/kg daily for 7 days, adjusting the volume of the freshly made solution and administering by intramuscular injection into the gluteal muscle, alternating sides each day.

Patients should preferably consume a sugar source (e.g., sweet drinks, biscuits, bananas, or other fruit) before injection to combat the common hypoglycemia reaction and lie down for at least 1 hour after injection to avoid symptomatic hypotension. During this time vital signs should also be monitored, particularly if the patient feels unwell.

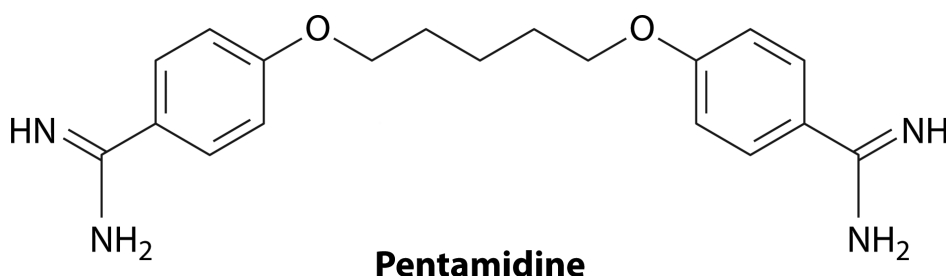


FIG 2 Structure of pentamidine.

Minor adverse reactions are common, although the drug is generally well tolerated. Immediate reactions may include hypotension in about 10% of patients, causing dizziness, collapse or shock (intravenous administration can lead to hypotension in up to 75% of cases), nausea, vomiting, and pain at the injection site, with the potential for sterile abscesses or necrosis occasionally occurring. Systemic effects can include azotemia due to nephrotoxicity, thrombocytopenia, leukopenia, anemia, elevated liver enzymes, and fluctuations in blood sugar levels. Rare but serious side effects include persistent diabetes, QT prolongation, arrhythmia, anaphylaxis, and acute pancreatitis.

Mode of action and resistance mechanisms

A definitive mode of action for pentamidine is still unknown. It binds DNA with high avidity within the minor groove of the double helix (67), and DNA binding has been proposed to account for at least some of the drug's activity.

Pentamidine is very potent against trypanosomes (IC₅₀ values *in vitro* falling in the 1–10 nM range). Time to kill depends on the dose, with increasing doses causing progressively faster killing (68), although removal of drug after a relatively short exposure still leaves parasites committed to death as a result of the drug rapidly reaching trypanocidal levels.

The drug reaches very high levels (in the low millimolar range) in trypanosomes exposed to low micromolar concentrations of the drug (69, 70). At least three plasma membrane transporters are involved in uptake (71). The original pentamidine transporter was identified as the TbAT1 (P2) aminopurine transporter (72). Its absence was proposed to relate to melarsoprol–pentamidine cross-resistance as melarsoprol too enters via the TbAT1 transporter (72, 73). However, other pentamidine transporters were also discovered, including one with exceptionally high affinity for drug (HAPT1) and another with lower affinity (LAPT1) (71). A genome-scale high throughput RNAi gene loss of function screen (74) aimed at seeking genes associated with resistance to trypanocidal drugs showed that TbAT1 was found in resistance to both melarsoprol and pentamidine (74). Another gene, *tbaqp2*, encoding TbAQP2, an aquaglyceroporin, was also identified as yielding melarsoprol and pentamidine resistance when its expression was diminished (74), and it was subsequently shown that TbAQP2 corresponded to the HAPT1 pentamidine transporter, which appears to be the primary pentamidine transporter (75, 76). In the field, in regions of the DRC where melarsoprol treatment failure was being reported in the early part of this century, TbAQP2 mutations were found (77) indicating that melarsoprol resistance was really becoming a problem prior to its replacement in stage 2 gambiense HAT treatment with eflornithine and subsequently NECT.

The mitochondrion has been proposed as a key target for pentamidine in various species including yeast (78). In *Leishmania* parasites (which are phylogenetically related to trypanosomes), mitochondrial damage (79) precedes cell death, and resistance in *Leishmania* correlates to a reduction in the mitochondrial membrane potential (80). Mitochondrial DNA in trypanosomes comprises the kinetoplast, an intercatenated network of circular DNA molecules (81) which is targeted by pentamidine. However, dyskinetoplastic bloodstream form *T. brucei*, which have lost or a disrupted kinetoplast, are only marginally less sensitive than wild-type cells to diamidines, indicating other targets. Fluorescent diamidines, such as furamidine (82), can be seen to accumulate rapidly in the kinetoplast and also in the acidocalcisome, another class of organelle involved in pH and cation homeostasis (83).

Within the human host, pentamidine has a large volume of distribution and a very long terminal half-life (the drug remaining detectable for many weeks after completion of a course) due to substantial protein binding and tissue retention. The long half-life made pentamidine a successful prophylactic in campaigns in central and west Africa in the mid-20th century (84), although the practice is no longer used.

NIFURTIMOX–EFLORNITHINE COMBINATION THERAPY

Background

NECT (Fig. 3) is highly effective for treating second-stage gambiense HAT (85), with cure rates between 95% and 98% with few side effects. Prior to 2009, eflornithine was used as monotherapy for stage 2 gambiense disease, replacing the highly toxic melarsoprol (86), which was retained only for stage 2 rhodesiense disease. The combination therapy is less likely to lead to drug resistance as well as reducing the total amount of eflornithine required and the dosing period. NECT remains the preferred treatment for stage 2 gambiense HAT with over 100 WBC/ μ L in CSF, i.e., late stage 2. It is also used for children under 6 years old or weighing less than 20 kg with second-stage gambiense HAT (more than 5 WBC/ μ L in CSF) where fexinidazole is not approved for use (27).

Clinical use

NECT is provided as a kit by the World Health Organization, which includes both drugs and other supplies for treating four adults (eflornithine hydrochloride [Ornidyl] in 50 mL vials [200 mg/mL] and nifurtimox [Lampit] as 120 mg tablets in 100 tablet bottles plus sterile water for injection in 250 mL plastic bags, syringes, and other accessories) (27).

The nifurtimox component is given at 15 mg/kg per day, taken in three oral doses (5 mg/kg every 8 hours) over 10 days (27). Tablets can be cut or crushed and mixed with food or sugary water if necessary, preferably under supervision, repeating the dose if vomiting occurs within 30 minutes. Eflornithine is administered at 400 mg/kg per day, given intravenously in two 2-hour infusions, dissolved in water for injection, or, if not available, in 5% dextrose or normal saline (0.9%) for 7 days. For children under 10 kg, the eflornithine is injected after dissolving in 50 mL of water for injection.

Adverse effects are similar to those of eflornithine and nifurtimox monotherapies, albeit generally better tolerated. Common reactions include abdominal pain, nausea, vomiting, and headache. Seizures, psychotic reactions, and hallucinations are possible. Diarrhea and vomiting are frequent but do not usually require stopping treatment. Other side effects include bone marrow suppression, tremor, headache, and vertigo.

Eflornithine can still be used as a monotherapy for second-stage gambiense HAT when NECT is not possible due to the unavailability or contraindications of nifurtimox and when fexinidazole cannot be used (27). As a monotherapy, eflornithine is given as a slow intravenous infusion of 400 mg/kg/day divided into four 6-hourly doses over 14 days (56 infusions in total).

Mode of action and resistance mechanisms

Eflornithine was designed as an inhibitor of the enzyme ornithine decarboxylase (ODC), a key enzyme of the polyamine biosynthetic pathway (87), with cancer therapy as the initial stimulus for development. The drug is an irreversible “suicide” inhibitor of the enzyme, meaning that as the enzyme recognizes the drug (an analog of the natural substrate, ornithine), it triggers a chemical reaction leading to the drug binding covalently to a cysteine residue (Cys360) in the enzyme's active site (88).

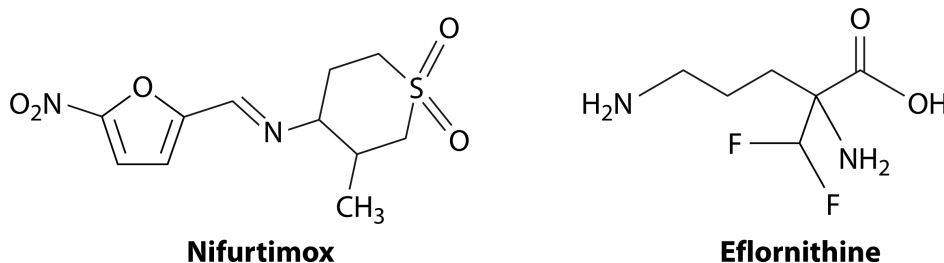


FIG 3 Structures of nifurtimox and eflornithine.

The compound is as potent against mammalian ODC as it is the trypanosome's enzyme (89). However, a crucial difference relates to the rate at which fresh enzyme is synthesized. In mammals, drug is rapidly turned over and replenished rapidly, with a half-life of around 30 minutes (90). The *T.b. gambiense* version, on the other hand, is only turned over every 12–18 hours (90). A corollary of this is that when trypanosomes are exposed to eflornithine, polyamine synthesis is inhibited and remains so for a protracted period. In mammalian cells, by contrast, inhibition occurs, but as drug levels wane between doses, enzyme is replenished, and polyamine synthesis resumes. What is more, trypanosomes have no plasma membrane transporters for polyamines and thus cannot accumulate extracellular polyamines from host body fluids. Many mammalian cells, however, can transport polyamines from their surrounding fluids also helping maintain polyamine levels when synthesis is inhibited (91).

Eflornithine is primarily trypanostatic and not trypanocidal at doses achieved in blood, requiring a functioning immune system for effective treatment, with the immune system destroying non-replicative (and thus non-antigenically varying) parasites.

It has been proposed that eflornithine's lesser activity against *T. b. rhodesiense* *in vivo* (92, 93) may relate to the faster turnover rate of ODC in rhodesiense, measured at 4.3 hours compared to 18 hours in gambiense (90). Another explanation, however, could relate to the relatively rapid progression of parasitemia in rhodesiense disease, which may make the trypanosome-induced immunosuppression (94) more pronounced.

Eflornithine is an amino acid, and it enters trypanosomes via the amino acid transporter, TbAAT6 (95–97), which can also carry other neutral amino acids with low affinity (96). Loss of the transporter leads to high-level resistance to eflornithine without negative impact on the parasites as either bloodstream forms or procyclic forms (98) since other transporters can carry the amino acid substrates of TbAAT6. Although several instances of laboratory selection for eflornithine resistance have all revealed loss of TbAAT6 in resistance (95–97), examples of resistance to the drug in a clinical setting have not yet been reported.

Nifurtimox is a nitrofurantoin and is activated by the same mitochondrial nitroreductase that also activates fexinidazole (51), and cross-resistance between these two nitroheterocycles is possible (54). In the case of nifurtimox, the product of TbNTR1-related reduction is an open-chain nitrile (51) believed to induce toxicity including through DNA damage. No indication of a synergistic effect between nifurtimox and eflornithine was noted in metabolomics analysis (99), and cytology-based profiling indicated a loss of mitochondrial membrane potential and diminished expression of mitochondrial proteins (100), indicating the mitochondrion as a main target.

SURAMIN

Background

Suramin (Fig. 4) is a polysulfonated symmetrical naphthalene derivative that was initially employed against sleeping sickness over 100 years ago (101), in 1922, following Ehrlich's discovery of the trypanocidal activity of naphthalene dyes such as trypan blue (102). Today, Bayer donates suramin to the WHO. In recent years, suramin's use has been mainly in treating stage 1 rhodesiense cases (27).

Clinical use

Bayer supplies suramin in vials containing 1 g of suramin sodium powder. This is reconstituted with 10 mL of sterile water prior to injection. Injection should be slow and intravenous (local irritation occurs if given intramuscularly (103)). Different dosing schedules are used in different national programs, all starting with a test dose in case patients are prone to rare (around 1 per 5,000 treated) hypersensitivity reactions which include urticaria and circulatory collapse. The drug should not be given in patients with severe renal disease and should be administered under close supervision, preferably with appropriate interventions at hand in case of anaphylaxis or cardiovascular adverse

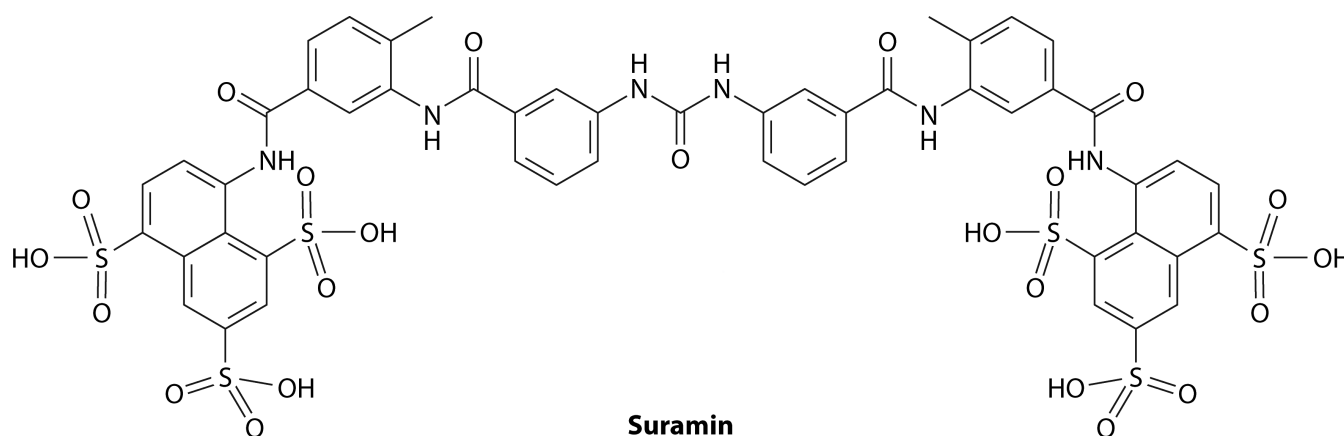


FIG 4 Structure of suramin.

reactions. A common suramin dosage regime consists of a test dose of 4–5 mg/kg on day 1, followed by five further injections of 20 mg/kg (maximum 1 g), diluted in water for injection, given weekly for 5 weeks (e.g., days 1, 8, 15, 22, and 29 after the test dose) (27). Suramin binds extensively (>99%) to plasma proteins contributing to a very long half-life. After a full course of 5–6 doses, the half-life is 44–54 days.

Pyrexia, nephrotoxicity, peripheral neuropathy, agranulocytosis, and thrombocytopenia are all reported adverse drug reactions along with the rare hypersensitivity responses noted above. Adverse events are relatively common, although generally mild and reversible.

Mode of action and resistance mechanisms

Suramin's precise mechanism of trypanocidal action remains elusive. It exhibits electrostatic interactions with numerous enzymes due to the negative charge on each of its six sulfonate groups, and scores of enzymes have been shown to be inhibited by the drug (101). Suramin's activity is significantly higher against bloodstream forms compared to procyclic forms, which could suggest glycolysis inhibition as a potential target (104) given the absolute dependence of the bloodstream form on glycolysis (59). Moreover, in a comprehensive investigation into changes to the metabolome and proteome associated with treating the parasites with suramin, a gradual loss of ATP production and switching on of the tricarboxylic acid (TCA) cycle and other aspects of mitochondrial metabolism normally suppressed in bloodstream forms was noted (105). Over 48 hours, the proteome also changed to a state where numerous enzymes associated with stumpy- or procyclic-form parasites were expressed (105). This raises the possibility that treatment with suramin could actually trigger differentiation pathways within the parasites within the bloodstream. This would be consistent with morphological changes noted after treatment (100), and cytokinesis defects have been reported in suramin-treated trypanosomes. Differentiation to stumpy forms also involves cell cycle arrest and loss of replication. Interestingly, suramin leads to parasite death more quickly in immunocompetent mice than in immunosuppressed mice indicating a role for the immune system (106). An induced differentiation state as proposed (105) with concomitant loss of replication could impact upon the surface integrity of the parasites making them more vulnerable to immune attack.

Suramin has been known to enter trypanosomes via receptor-mediated endocytosis for some time (104), and this could also contribute to its enhanced activity against bloodstream forms over procyclic forms. The pivotal genome-wide RNAi screen for resistance determinants (74) identified ISG75 as a likely receptor for suramin, although parasites lacking ISG75 retain substantial sensitivity to suramin (107), which points to

additional receptors involved in uptake too, with low-density lipoprotein receptors of a currently unknown molecular identity being suspected (108).

The RNAi screen (74) also demonstrated that an endosomal transporter of the major facilitator superfamily played a role in the overall action of suramin since its loss was associated with resistance, indicating a possible role in the release of suramin from endocytic vesicles into the cytosol where its action might begin to be felt.

Reports of suramin resistance in the field are rare, but resistance is readily achievable in laboratory settings (74). Selection of high-level suramin resistance in one *T. brucei rhodesiense* line *in vitro* (109) led to the identification of parasites expressing a particular VSG variant (VSG^{sur}) that bound suramin and led to its endocytosis via a route different from that associated with ISG75 or low density lipoprotein (LDL) uptake, which in turn probably prevented its reaching its intracellular target(s). *In vivo*, such parasites would be unlikely to survive since an immune response against VSG^{sur} would destroy these parasites once established, and any switching to another VSG would regain suramin sensitivity. Additional resistance was given to those lines through mutations to a RuvBL1-like DNA helicase (110). A point mutation found in the resistant parasites's version, when engineered into the gene in wild-type parasites, yielded high-level resistance, and *T. evansi* resistant to suramin also had that same amino acid in the cognate position (110).

MELARSOPROL

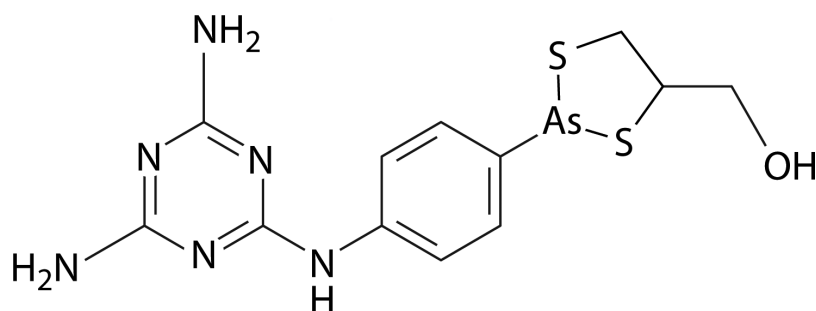
Background

Melarsoprol (Fig. 5) is a melaminophenyl-based organic arsenical introduced in 1949 (111) as a replacement of the previously used arsenical, tryparsamide, which had extremely high rates of toxicity including induction of blindness in many patients (112). Although preferable to tryparsamide, melarsoprol's use is marred by serious adverse effects, with approximately 5%–10% of patients experiencing reactive encephalopathy, a condition associated with a mortality rate as high as 50% (27). The compound comprises melarsen oxide linked to British anti-Lewisite, which stabilizes the arsenical, although melarsen oxide is rapidly regenerated once the compound enters aqueous solution (113).

Melarsoprol was gradually replaced, initially by eflornithine and then, in 2009, by NECT for treatment of gambiense disease but remained recommended for stage 2 rhodesiense HAT until the recent inclusion of fexinidazole for this condition in WHO guidelines (27). The only remaining indication of melarsoprol is in children aged <6 years or body weight <20 kg who are in second-stage rhodesiense HAT and in other rhodesiense HAT patients in second-stage who cannot receive fexinidazole for any reason. If acoziborole also becomes available for stage 2 rhodesiense disease, melarsoprol can be withdrawn without hindrance to any disease ameliorative activities.

Clinical use

The drug, trade-named arsobal, is provided by Sanofi in 5 mL ampoules containing 180 mg of melarsoprol dissolved in propylene glycol (36 mg/mL [3.6%]) given its insolubility in water. The propylene glycol solution is preferentially given in sterile dry glass syringes, provided appropriate sterilization procedures are available since many common plastics are sensitive to this solvent. Melarsoprol is given at a dose of 2.2 mg/kg once per day (in a maximum of 5 mL of the product) by slow intravenous injection for 10 days (27). This dosing was chosen following careful pharmacokinetic analysis (114), while the previous dosing had been based on earlier experience with tryparsamide which required a long regimen involving protracted rest periods between injections to prevent toxic tissue accumulation. The introduction of the 10-day regimen marked the beginning of an evidence-based approach to trypanocide regimen setting and gave advantages such as reducing the time required for hospitalization of patients (115). Melarsoprol is preferentially administered with the anti-inflammatory drug prednisolone (1 mg/kg/day



Melarsoprol

FIG 5 Structure of melarsoprol.

to a maximum dose 50 mg for 9 days, tapering the dosage: 0.75 mg/kg, 0.5 mg/kg, and 0.25 mg/kg, on days 10–12 (116). Melarsoprol injection is painful, and each daily injection should be given into a fresh site by experienced healthcare personnel.

Adverse effects of melarsoprol culminate in an encephalopathic syndrome that occurs in 5%–18% of all treated cases and is fatal in 10%–70% of these patients (117). The encephalopathy usually occurs 3–10 days after the first treatment. Fever and convulsion are the first signs of the response, and these progress to further neurological disorders, behavioral changes, progressive coma, and death. Patients should be monitored, and at the first sign of potential encephalopathy, further melarsoprol injections should be stopped, and dexamethasone and diazepam may be considered to dampen any inflammatory response. Other adverse reactions are common, including gastrointestinal issues (nausea, vomiting, and diarrhea) and skin reactions (pruritus and exfoliative dermatitis), albuminuria, hypertension, and myocardial damage. Cardiac failure is also common and can be fatal.

Modes of action and resistance mechanisms

The precise mechanisms underlying the trypanocidal activity of melarsoprol/melarsen oxide are unknown. *T. brucei* undergoes rapid lysis when exposed to relatively high levels of melarsoprol/melarsen oxide *in vitro* (118). Exactly why though is not clear, although loss of ATP and its interactions with cellular thiols, including the trypanosomatid specific polyamine–glutathione compound named trypanothione (119), have been proposed as key. The drug has also been shown to prevent mitosis (100), possibly through specific mitogen activated protein (MAP) kinase-dependent pathways. Melarsoprol also inhibits DNA synthesis in a manner that can be reversed by over-expressing glutathione (120), thus creating more trypanothione, a known target of the drug and key cofactor in deoxynucleotide synthesis through ribonucleotide reductase. A degree of polypharmacology is thus key in the drug's mode of action, and different doses yield different times to kill, perhaps reflecting the different targets hit. Fifty-seven genes, when over-expressed, yielded some resistance to the drug in a genome-scale screen (120), further pointing to a complex interaction between drug and parasite. In rodent models of the disease, a functional immune system is required to achieve sterile cure following melarsoprol treatment (106), echoing the situation with eflornithine and suramin where trypanostatic effects that influence cell surface architecture and the ability to evade the immune system are compromised. Laboratory studies initially revealed resistance to be associated with loss of the P2 aminopurine transporter (72), which can also carry melaminophenyl arsenical drugs as well as diamidines. The transporter (also known as TbAT1) was later shown to be encoded by the *tbat1* gene (73). Subsequently, it was shown that the aquaglyceroporin TbAQP2 (74), which corresponds to the high-affinity pentamidine transporter (HAPT1) (71), also carries melarsen oxide. The emergence of treatment failures with melarsoprol in the DRC in the early 21st century (77) was shown

to correspond to the appearance of mutant alleles of the *tbaqp2* gene, indicating that resistance was a growing problem when the switch to NECT therapy was made. Resistance can also come about through over-expression of the *tbmprpa* gene encoding the multidrug resistance protein A (MRPA) efflux pump that ultimately removes melarsenoxide bound to trypanothione from the cell (121).

Efforts to improve the drug's safety profile have explored conjugating melarsoprol into complexes to potentially slow the release of its active components, thus reducing toxicity (122); however, the advances in other areas of HAT therapy mean such a product is unlikely to be required, and melarsoprol's use may be avoidable in the future.

ACOZIBOROLE

Background

Acoziborole (Fig. 6) belongs to the benzoxaborole class of compounds which have shown considerable promise against a range of pathogenic microorganisms (123, 124). The class was pursued by Anacor pharmaceuticals (who were purchased by Pfizer Inc. in 2016), and benzoxaborole activity against trypanosomes was first reported in 2010 (125). This led to DNDi partnering with Scynexis Inc. to develop the series toward potent trypanocides with pharmacokinetic properties enabling penetration of the CNS, rendering them suitable to cure stage 2 HAT. A 6-carboxamido-based series was shown to cure the mouse model of stage 2 disease (126), and further medicinal chemistry created SCYX-7158 (AN5568) which cured stage 2 mouse models even when given orally at a daily dose of 25 mg/kg for 7 days (127). Preclinical testing in dogs and rodents revealed good safety characteristics (125), and in 2012, the compound entered clinical trials for HAT.

One hundred twenty-eight healthy male subjects of sub-Saharan African origin were included in an initial phase 1 safety study which included assessing tolerability, pharmacokinetics, and pharmacodynamics after a single oral, ascending dose (128). Adverse effects including headaches, dizziness, and GI tract reactions (e.g., abdominal pain, nausea, vomiting, constipation, and diarrhea) were reported, but all were mild, and serious issues were not identified.

This was followed by a multicenter phase 2/3 study involving 208 gambiense HAT patients aged 15 years or older (167 at stage 2 and 41 stage 1 or intermediate stage) in sites across the Democratic Republic of the Congo and Guinea (129). Oral acoziborole was administered as a single 960 mg dose (3×320 mg tablets) to fasted patients who were then observed for 18 months post-treatment at which point the primary efficacy endpoint was measured as acoziborole cure rate. All stage 1/intermediate stage patients were cured (100% success rate), while for stage 2 patients, the cure rate was 95% (better than the historical cure rate of NECT at 94%). Side effects were minimal with pyrexia and asthenia most common. Final considerations on the manufacture and administration of the drug are under review and additional trials, e.g., testing efficacy and dosing in children are underway. The prospect of a highly efficacious drug, effective against stage 2 disease given in a single dose, is truly remarkable and offers opportunities to revolutionize HAT treatment. For example, a "screen and treat" program could be implemented, where even relatively insensitive screening tools could be used to determine whether there is a suspicion of HAT in people within a particular locality. For those in whom there is such a suspicion, acoziborole could be administered without concern for potential adverse events, if further safety data permits, and without having to differentiate stage 1 vs stage 2.

Mode of action and resistance mechanisms

Understanding of the mode of action and resistance mechanisms to benzoxaboroles is emerging. Selection of resistance by growing bloodstream-form trypanosomes in ascending doses of drug led to the identification of multiple genetic changes in trypanosomes (130) including gene copy number amplification of the RNA cleavage

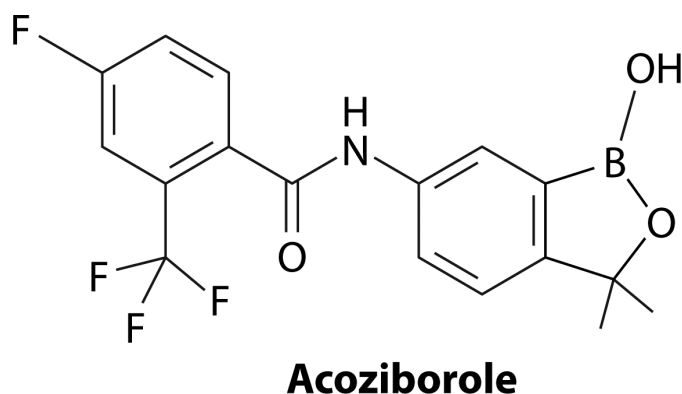


FIG 6 Structure of acoziborole.

and polyadenylation specificity factor subunit 3 (CPSF3). CPSF3 was later identified as a benzoxaborole target in *Plasmodium* (131) and *Toxoplasma* (132). Metabolomics experiments in *T. brucei* had revealed a profound change in the methionine metabolism (133) indicative of RNA processing defects, and over-expression of CPSF3 was shown to diminish sensitivity to the related benzoxaborole, AN7973 (134). A genome-scale approach to the problem revealed independently that over-expression of CPSF3 caused resistance (135). Another series of experiments selecting resistance in *in vitro* culture yielded an interesting result where it appeared that bloodstream-form parasites, exposed to the drug and selected for resistance, had altered their gene expression profile to resemble stumpy or procyclic forms (136). Assessment of the sensitivity of procyclic forms to acoziborole revealed them to be significantly (~7.5-fold) less sensitive to the drug (136). A model whereby the mode of action simply involves inhibiting the CPSF3 target seems difficult to reconcile, as the protein should play similar roles in both forms of the parasite. However, a genome-scale RNAi screen on bloodstream forms pointed to a significant role in the machinery involved in protein SUMOylation as affecting sensitivity (parasites in which genes encoding different elements of the machinery were downregulated showed very high levels of resistance to the drug [137]). It was shown that CPSF3 bound to acoziborole was SUMOylated much more readily than the enzyme without drug bound. Thus, it is possible that the drug works by promoting degradation of the CPSF3 protein following its SUMOylation which is promoted by binding drug (137). Interestingly, it was recently shown that SUMOylation is intimately related with slender to stumpy differentiation (138); thus, the resistance phenotype reported (136) could involve a primary impact on SUMOylation that yielded resistance to drug but also triggered differentiation. Intriguingly, *Plasmodium* parasites also develop high-level resistance to a CPSF3-targeting benzoxaborole by disrupting SUMOylation (139), pointing to conservation of this novel action mechanism.

FURTHER DEVELOPMENTS

High throughput screening and the identification of a rich vein of new trypanocides

Modern drug discovery has benefitted from the generation of enormous compound library collections and, critically, the ability to screen these against potential drug targets, or against cellular systems, including parasites (140). The former approach is often referred to as target-based drug discovery (141), while the latter has been referred to as phenotypic screening (142), since it is an impact on the phenotype of a cellular system that is measured. In the case of trypanosomes, the phenotype is simply to measure cell death. A number of medium to high throughput screens have been conducted, and several very promising compounds that are active against the parasites and cure rodent models of disease have been identified. An early example (143) involved screening a collection of 2,160 FDA-approved drugs used at 1 μ M against bloodstream-form

trypanosomes using an easy-to-use ATP-based bioluminescence assay. An advantage of finding hits in this particular screen would be that any compounds showing suitable activity could, in principle, be further developed relatively quickly given they were already approved for use in humans and had substantial accompanying pharmacological information. Among the 35 compounds identified as trypanocidal in the assay were suramin and pentamidine. Among others, a variety of anticancer cytotoxins and other antimicrobial agents were identified. None were pursued, but the work revealed the potential to screen large compound libraries.

Another key advance came with the screening of N-myristoyltransferase (NMT) (144), an enzyme that had previously been shown to be essential to bloodstream-form trypanosomes and thus a possible drug target. Screening for NMT inhibitors among a library of 62,000 chemically diverse compounds identified the drug-like pyrazole sulfonamide scaffold among the good inhibitors (144). From the initial hit, a series of over 200 derivatives were synthesized and tested for activity against the enzyme (using mammalian NMT as a counter screen to assure selectivity). Compounds were then tested for their ability to kill bloodstream-form trypanosomes with selectivity over mammalian cells, and additional modifications to the core structure were made to optimize activity against a mouse model of trypanosomiasis. Highly potent trypanocides were found; however, it proved difficult to develop derivatives that could cross the blood–brain barrier to reach therapeutic concentrations in the brain (144), thus the series never progressed to clinical development for HAT (145).

Two pharmaceutical companies, in particular, embraced the quest for new drugs against the diseases caused by kinetoplastid parasites. Glaxo Smith Kline (GSK) established their Open Lab in Tres Cantos and screened a 1.8-million compound library (146) against *T. brucei* and also *Leishmania donovani* and *T. cruzi*. The *T. brucei* screen was focused on bloodstream form organisms *in vitro* using a resazurin-based assay. 27,600 compounds were found to be active at the starting concentration of 4.2 μ M, of which 4,200 were taken further for dose-response studies, selecting compounds whose physicochemical characteristics were most likely to yield good drug-like molecules with CSF penetration. After counter-screening against mammalian cells, a total of 700 compounds remained. The GSK hits have underpinned further development against *Leishmania* (147, 148), although relatively little further work was conducted toward *T. brucei*, as fexinidazole and acoziborole were already progressing toward clinical development at the time of the screen.

At around the same time, the Novartis Institute of Tropical Disease performed analogous screens of their 3-million compound library collection (149) against *T. brucei*, *T. cruzi*, and *L. donovani*, hoping to find compounds that could be further developed as pan-kinetoplastid drugs. The *T. brucei* screens used a similar ATP-detecting assay as that mentioned above. The screen identified a series of promising compounds including inhibitors of the trypanosomatid proteasome (149) (and one derivative, LXE408 (Fig. 7), has now entered clinical trials against *Leishmania* (150) and also pre-clinical testing against Chagas disease). Having identified the proteasome as a target, Novartis rescreened the library in a fluorescence-based assay against a *Leishmania tarentolae* version of the 20 S proteasome and identified further inhibitors including a series based around a quinoline chemotype (151). Cryo-electron microscopy (CryoEM) structures of compounds bound to the proteasome enabled adaptations to create better inhibitors, and trypanocidal derivatives were found, including some that had good brain penetration and the ability to cure a rodent model of stage 2 disease (151). The same original phenotypic screen (149) also formed the basis of further work that has led to the discovery of highly potent inhibitors of the kinetochore-associated kinase CLK1 (152) and also a series of potent cyanotriazole compounds (Fig. 7) that inhibit trypanosomatid topoisomerase IIa and can cure mouse models of stage 2 disease (153). Currently, a number of these compounds continue toward clinical development against *Leishmania*-s and Chagas disease, but their future development for HAT remains on hold given the successes elsewhere. However, the identification of a number of potent trypanocides

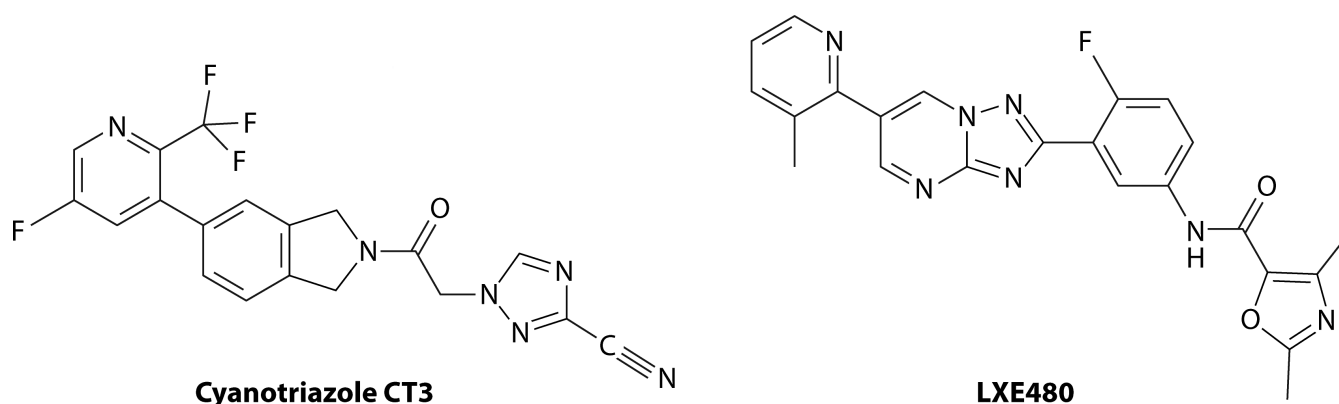


FIG 7 Structures of cyanotriazole CT3 and LXE480.

with excellent therapeutic potential within a relatively short time frame over the first part of the 21st century represents a truly transformative situation with respect to the treatment of HAT.

CONCLUSIONS

With regard to sleeping sickness, the 21st century began tragically much as the 20th century had, with epidemics running out of control across many places in sub-Saharan Africa. The situation was ridiculous given that in 1900, it had not been shown that the trypanosome was responsible for the disease, nor the tsetse fly responsible for its transmission. There were no drugs or other interventions. By 2000, it was known that two sub-species of trypanosome, *T. b. gambiense* and *T. b. rhodesiense*, cause the disease, and treatments were available to treat both types of disease at both stages (stage 1 before the parasites were manifesting overt disease in the CNS and stage 2, after that point). For the most part, however, those drugs were archaic remnants of an age when generally poor compounds, provided they had some benefit beyond the adverse events they caused, could be taken forward for clinical use. For HAT, the melaminophenyl arsenical, melarsoprol, the most widely used trypanocide at the turn of the century was a case in point, killing in the order of 5% of those treated. The advent of a new model of socio-economic responsibility in drug development has seen remarkable progress in the past decade. Spearheaded through the World Health Organization, national sleeping sickness programs were revitalized. New diagnostic tools have appeared, and critically, new and better medicines have become available. Initially through carefully revising regimens of existing drug (e.g., shifting melarsoprol to a 10-day regimen), then expanding the use and improving eflornithine through combining with nifurtimox. The foundation of new product development partnerships, such as the Consortium for Parasitic Drug Development and DNDi, offered a new model to discover and develop medicines for diseases of the world's most marginalized communities. DNDi, in particular, galvanized new ways of seeking investment to develop new compounds. This led to the development of fexinidazole, an orally available drug now used for the vast majority of all trypanosomiasis cases. Fexinidazole, however, needs to be taken with food and over 10 days. Another orally available compound, acoziborole, has now been shown to be efficacious and can cure stage 2 gambiense disease, as well as stage 1, in just a single administration. Current safety data are promising, and this could open the way to a campaign of screen and treat where even relatively ineffective diagnostic tests can point to clinical suspects and allow their safe treatment. In the meantime, diagnostics are improving, and control of the tsetse fly vectors that transmit the disease has become more tractable too. It appears, therefore, that the WHO goal of eliminating transmission of gambiense HAT is achievable. The chemotherapy of human African trypanosomiasis has truly been transformed.

AUTHOR AFFILIATION

¹School of Infection and Immunity, College of Medical, Veterinary and Life Sciences, University of Glasgow, Glasgow, United Kingdom

AUTHOR ORCIDS

Michael P. Barrett  <http://orcid.org/0000-0001-9447-3519>

AUTHOR CONTRIBUTIONS

Michael P. Barrett, Conceptualization, Investigation, Project administration, Writing – original draft, Writing – review and editing

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AUTHOR BIO

Michael P. Barrett is a professor of Biochemical Parasitology at The University of Glasgow's School of Infection & Immunity. His work focuses on understanding how drugs work against parasitic protozoa, particularly trypanosomes and leishmania, and the processes by which these parasites evolve resistance to drugs. Barrett is part of the World Health Organisation's expert committee on human African trypanosomiasis and part of the Drugs for Neglected Diseases initiative Scientific Advisory Committee.

